deep-learningexp1.ipynb

[1]
```python
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from sklearn.preprocessing import OneHotEncoder
```

[2]
```python
(X_train, y_train), (_, _) = mnist.load_data()
X_train = X_train.reshape(-1, 784) / 255.0
encoder = OneHotEncoder(sparse_output=False)
y_train = encoder.fit_transform(y_train.reshape(-1, 1))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

[3]
```python
input_size, hidden_size, output_size = 784, 64, 10
W1 = np.random.randn(input_size, hidden_size) * 0.01
b1 = np.zeros((1, hidden_size))
W2 = np.random.randn(hidden_size, output_size) * 0.01
b2 = np.zeros((1, output_size))
```

[4]
```python
sigmoid = lambda x: 1 / (1 + np.exp(-x))
sigmoid_deriv = lambda x: x * (1 - x)
loss_fn = lambda y, y_hat: -np.mean(y * np.log(y_hat + 1e-8))
```

[9]
```python
epochs, lr = 10, 0.1
losses = []
for epoch in range(epochs):
  total_loss = 0
  for i in range(X_train.shape[0]):
      x = X_train[i:i+1]
      y = y_train[i:i+1]
      z1 = x @ W1 + b1
      a1 = sigmoid(z1)
      z2 = a1 @ W2 + b2
      a2 = sigmoid(z2)
      loss = loss_fn(y, a2)
      total_loss += loss
      dz2 = a2 - y
```

Classroom Management Tools | lab manual-deep learning - Cl | ChatGPT | deep-learningexp1.ipynb - Col | Upload files · Srenith19/Deep-

colab.research.google.com/drive/1A3Fr_wa2lqoiwShpEa13xT8E88Ila5M2#scrollTo=f95RzkKQXVlq

deep-learningexp1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
Epoch 7, Loss: 0.0087
Epoch 8, Loss: 0.0078
Epoch 9, Loss: 0.0075
Epoch 10, Loss: 0.0070
```

[10]

```
plt.plot(losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()
```

[11]

```
def predict(img):
    img = img.reshape(1, 784) / 255.0
    a1 = sigmoid(img @ W1 + b1)
```

Variables Terminal

13:54 Python 3

Classroom Management Tools | lab manual-deep learning - Cl | ChatGPT | deep-learningexp1.ipynb - Col | Upload files · Srenith19/Deep-

colab.research.google.com/drive/1A3Fr_wa2lqoiwShpEa13xT8EB8JIa5M2#scrollTo=f95RzkKQXVlq

deep-learningexp1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[9]

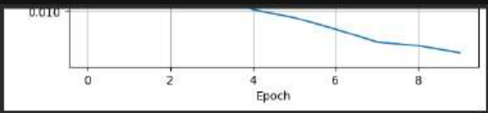
```python
losses = []
for epoch in range(epochs):
  total_loss = 0
  for i in range(X_train.shape[0]):
      x = X_train[i:i+1]
      y = y_train[i:i+1]
      z1 = x @ W1 + b1
      a1 = sigmoid(z1)
      z2 = a1 @ W2 + b2
      a2 = sigmoid(z2)
      loss = loss_fn(y, a2)
      total_loss += loss
      dz2 = a2 - y
      dW2 = a1.T @ dz2
      db2 = dz2
      dz1 = (dz2 @ W2.T) * sigmoid_deriv(a1)
      dW1 = x.T @ dz1
      db1 = dz1
      W2 -= lr * dW2
      b2 -= lr * db2
      W1 -= lr * dW1
      b1 -= lr * db1
  losses.append(total_loss / X_train.shape[0])
  print(f"Epoch {epoch+1}, Loss: {losses[-1]:.4f}")
```

```
Epoch 1, Loss: 0.0275
Epoch 2, Loss: 0.0159
Epoch 3, Loss: 0.0130
Epoch 4, Loss: 0.0114
Epoch 5, Loss: 0.0101
Epoch 6, Loss: 0.0096
Epoch 7, Loss: 0.0087
Epoch 8, Loss: 0.0078
Epoch 9, Loss: 0.0075
Epoch 10, Loss: 0.0070
```

[10]

```python
plt.plot(losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
```

Variables Terminal 13:54 Python 3




```python
def predict(img):
    img = img.reshape(1, 784) / 255.0
    a1 = sigmoid(img @ W1 + b1)
    a2 = sigmoid(a1 @ W2 + b2)
    return np.argmax(a2)
idx = 100
plt.imshow(X_train[idx].reshape(28, 28), cmap='gray')
plt.title(f"Prediction: {predict(X_train[idx])}")
plt.axis('off')
plt.show()
```